

Module 2 — Read 0

Project preview — Project 2 — Estimating the Truth

Module 2, Read 0: Why estimation matters

Big question: What is the truth about a **whole population**, and how close can we get from one **sample**?

Project: [Project 2 — Estimating the Truth](#) · **Sample report:** [CA high schools example \(PDF\)](#) · **Next:** [Read 1](#)

i Short read — start here

Five-minute orientation before [Read 1](#). Module 1 asked “*Is this observation surprising?*” Module 2 asks “*What is a plausible range for the population truth?*”

The story in one pass

In Module 1 you tested a **claim** with a p-value. In Module 2 you **estimate** an unknown **parameter** (population proportion p or mean μ) using a **statistic** from your sample ($\hat{p}$ or $\bar{x}$).

A **95% confidence interval** is a range of plausible values for the parameter. The interval width reflects **precision** — larger n usually means a narrower interval. **How** you sample matters too: a **simple random sample** from a good **sampling frame** supports generalizing to the population; convenience samples often do not.

Project 2 requires you to define a **finite, listable population** (N), draw an **SRS** of size n , compute the right **95% CI** (mean or proportion; **FPC** when sampling without replacement from finite N), and interpret it with the sentence pattern from Read 1.

Tour the sample project

Open the **sample one-page write-up — CA high schools (PDF)** and the [HTML example](#). **Do not copy that population** — build your own study.

Section	What to notice	Why it matters for <i>your</i> project
Introduction	Research question, population, variable type	You must name a listable finite population and one variable
Methods	How the SRS was drawn; n and N	Project 2 requires documented SRS from $\{1, \dots, N\}$
Results	Point estimate + 95% CI + conditions checked	Match formula to mean vs proportion and FPC when needed
Conclusion	“I am 95% confident that...” in context	Same interpretation template you will use on the rubric

What you will do (different from the sample)

1. Choose your **own** finite population and variable (binary $\rightarrow$ proportion, or numerical $\rightarrow$ mean).
2. Draw an **SRS** without replacement and collect real data.
3. Compute a **95% CI** with the correct formula and **check conditions**.
4. Submit a **one-page summary** + runnable R notebook ([mean template](#) or [proportion template](#)).

What to do next

1. Skim the **sample PDF** (5–10 minutes).
2. Open **Project 2** and note due dates on the **course path**.
3. Start **Read 1** — binomial center/spread through CIs and sampling.

When you finish Read 1, take **Quiz A (Def & Notation)**, then continue with **Read 2**.